Affordable Real Style Transfer: Training Wavelet-Decoupled Rectified Flow on a Single RTX 3060 in Minutes

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Abstract

Art-to-art style transfer is easy to overclaim because the source image is already an artwork. If the unchanged source already resembles the target domain, high style similarity can come from doing almost nothing. We formalize this issue with identical-image transfer (IDT). The unchanged source defines a no-op floor. A method below that floor has not moved in the requested direction.

We then analyze why compact flow models often fall into this regime. In Euclidean latent coordinates, low-frequency layout motion and high-frequency texture motion share one basis. The flow objective can then favor structure preservation and suppress style-conditioned transport. We state a sufficient local condition for this style-suppression regime and turn it into ablation-level predictions.

We propose WD-VF, a wavelet-decoupled rectified flow in the latent space of the Stable Diffusion v1.5 EMA VAE. A single-level Haar transform separates LL, LH, HL, and HH subbands. WD-VF weakens LL supervision, routes style queries through high-frequency bands, and applies whitening and coloring only at the endpoint. This keeps global structure stable while concentrating model capacity on stroke and texture transfer.

On Distinct5-WikiArt, the WD-VF model (with a trainable flow backbone of 903K parameters, excluding the frozen VAE) reaches CLIP-S = 0.7213 and LPIPS = 0.2868. It clears the IDT floor of 0.6933 and exceeds SaMam by +0.1397 CLIP-S. Relative to Seedream 4.5, it reduces LPIPS by 39.8% while staying in the same ArtFID range. Training takes 3.08 minutes on a single RTX 3060 (12 GB). Generation plus VAE decode takes 83.7 seconds for 750 outputs, or 111.7 ms/image. Furthermore, this decoupled geometry enables lightweight few-shot adaptation. Adding new styles requires updating only a compact style-memory module without retraining the core transport flow, preserving high-quality transfer. The ablations follow the predicted pattern: removing DWT weakens the frontier, route mismatch lowers transfer quality, and repeated style injection raises LPIPS.

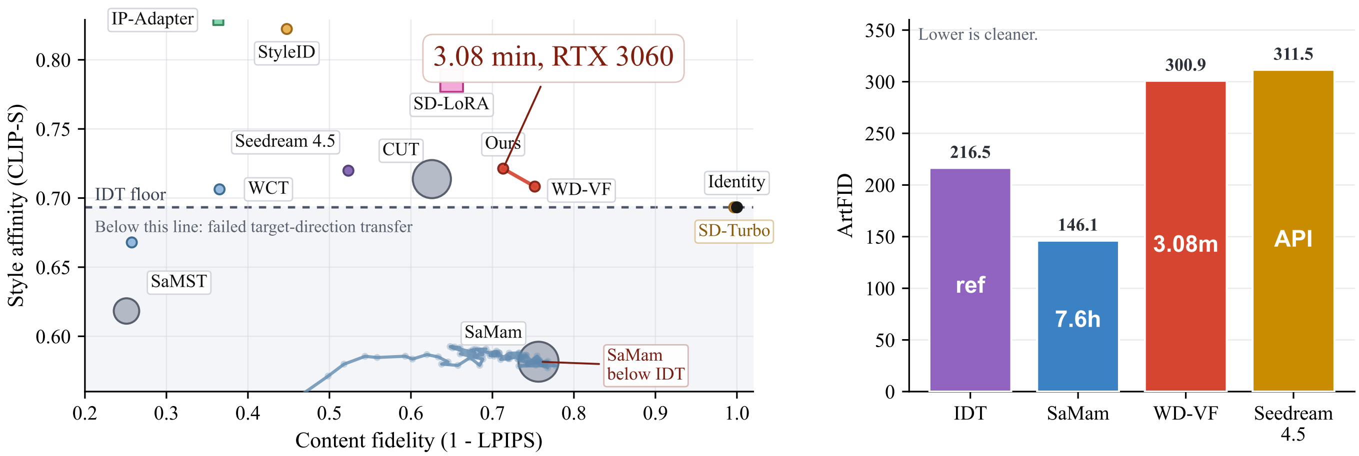

Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: CLIP-S versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WD-VF operating points. Square markers denote training-free diffusion editors (StyleAligned, IP-Adapter) and PEFT baselines (SD-LoRA); these are included to anchor the "efficiency–quality" trade-off. The dashed line marks the IDT floor (0.6933), which SaMam falls below. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

1 Introduction

Art-to-art style transfer has an unusually forgiving failure mode. The source image is already an artwork, so doing almost nothing can still produce a high style score. If the unchanged source already lies close to the target domain, raw CLIP similarity does not prove that the method moved in the requested direction. We therefore calibrate this setting with identical-image transfer (IDT). The unchanged source defines a no-op floor. Real transfer must clear that floor.

This calibration changes the ranking on Distinct5-WikiArt. We build the benchmark from the five WikiArt styles with the lowest IDT CLIP-S in a broader style pool. This makes accidental no-op success harder. Under the unified 750-pair protocol, SaMam reaches LPIPS 0.2434 yet only CLIP-S 0.5816. That score is far below the IDT floor of 0.6933. The result is not strong transfer with low damage. It is near-reconstruction in the wrong direction. This empirical signature—low LPIPS together with sub-IDT CLIP-S—is exactly what Theorem 1 predicts for a compact model trained in Euclidean latent coordinates: the flow loss pulls the field toward the style-suppression manifold $\mathcal{M}_0$.

We trace this failure to the geometry of Euclidean latent flow. In standard latent coordinates, the rectified-flow target $u_t = z_1 - z_0$ mixes low-frequency layout motion with high-frequency texture motion. The flow loss can then reward structure preservation more strongly than style-conditioned motion. Optimization is pulled toward a style-suppression regime. Theorem 1 formalizes one sufficient local condition for that regime. The theorem also yields concrete predictions. Removing the coordinate change should weaken the CLIP-S/LPIPS frontier. Route mismatch should lower transfer quality. Repeated style injection should increase content damage.

WD-VF alleviates this issue by changing coordinates rather than enlarging the generator. It applies a single-level Haar transform, weakens LL supervision, routes style queries through the high-frequency bands, and stylizes only the endpoint subbands. In the ideal routed limit, the active supervised residual reduces to two orthogonal high-frequency bands. This removes the dominant low-frequency conflict from both the style-query path and most of the supervised transport.

The resulting model is explicit and small. It uses 903K trainable parameters in its flow backbone (excluding the frozen VAE). It reaches CLIP-S 0.7213 and LPIPS 0.2868, which clears the IDT floor. Among trained local baselines in this benchmark, it gives the strongest positive-IDT result. Training takes 3.08 minutes on a single RTX 3060. Generation plus VAE decode takes 83.7 seconds for 750 outputs, or 111.7 ms/image.

Contributions. (1) We introduce IDT-calibrated evaluation for art-to-art transfer. It distinguishes real target-direction transfer from no-op behavior on artwork inputs. (2) We analyze a local style-suppression regime in Euclidean latent flow. Theorem 1 states a sufficient curvature condition and yields ablation-level predictions. (3) We propose **WD-VF**, a wavelet-decoupled rectified flow with weak LL supervision, high-frequency query routing, and endpoint subband alignment. In the ideal routed limit, the active supervised transport reduces to two orthogonal high-frequency bands (Section 3.4). (4) We show that this geometry gives an efficient real-transfer regime. WD-VF uses 903K trainable parameters (excluding the frozen VAE) and trains in 3.08 minutes. Crucially, the decoupled transport is highly extensible: we demonstrate that the model can generalize to unseen styles via lightweight style-memory updating, eliminating the need for full retraining.

2 Related Work

Exemplar methods. Classical exemplar transfer uses a target image at inference time (Gatys, Ecker, and Bethge 2015, 2016; Huang and Belongie 2017; Li et al. 2017; Deng et al. 2022; Huang et al. 2024). These methods match target statistics or target correspondences. Our setting uses only the source image and a target style identity. The task is therefore domain-conditional translation rather than exemplar stylization.

Domain-conditional methods. The closest learned baselines are compact multi-style systems such as SaMST and SaMam (Liu et al. 2024a,b). Larger diffusion-based editors such as StyleID and SDEdit define the high-compute end of the same benchmark range (Meng et al. 2021; Chung, Hyun, and Heo 2024). Training-free diffusion editors such as StyleAligned (Gal et al. 2024) and IP-Adapter (Ye et al. 2023) offer zero-shot style control without retraining, while PEFT approaches such as SD-LoRA (Hu et al. 2022) fine-tune a small rank decomposition on a target style. Together with commercial systems such as Seedream (Zhang et al. 2024), these methods anchor the "quality" side of the efficiency–quality spectrum. WD-VF targets the same interface at much lower cost.

Wavelets and evaluation. Prior wavelet methods mainly use multi-scale decompositions as feature-processing modules (Daubechies 1992; Phung, Ghosh, and Tran 2023). We instead use a wavelet basis as a coordinate change that separates low-frequency structure from high-frequency style motion. We pair this design with IDT calibration because the source image is already an artwork. The main metrics are CLIP-S, LPIPS (Zhang et al. 2018), and the gap to the IDT floor. ArtFID (Wright and Ommer 2022) serves as an auxiliary check.

3 Theory and Method

3.1 Setup

Let $x_c \in \mathbb{R}^{H\times W\times 3}$ be a content image and $s \in \{1,\dots,K\}$ a target style domain. We work in the latent space of the Stable Diffusion v1.5 EMA VAE. We write

$$z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4\times 32\times 32}. \tag{1}$$

During training, $z_1 \sim p_s$ is a latent sampled from the target domain. Rectified flow uses the straight segment

$$\begin{aligned} z_t &= (1-t)z_0 + t z_1, \\ u_t &= z_1 - z_0, \qquad t \sim \mathcal{U}([0,1]). \end{aligned} \tag{2}$$

At inference we use an 8-step Heun solver (Heun gave the best cost-quality tradeoff in our controls; Euler loses style, RK4 adds cost without improving the final result). We then decode $\hat{x}_s = \mathcal{D}(\hat{z}_1)$.

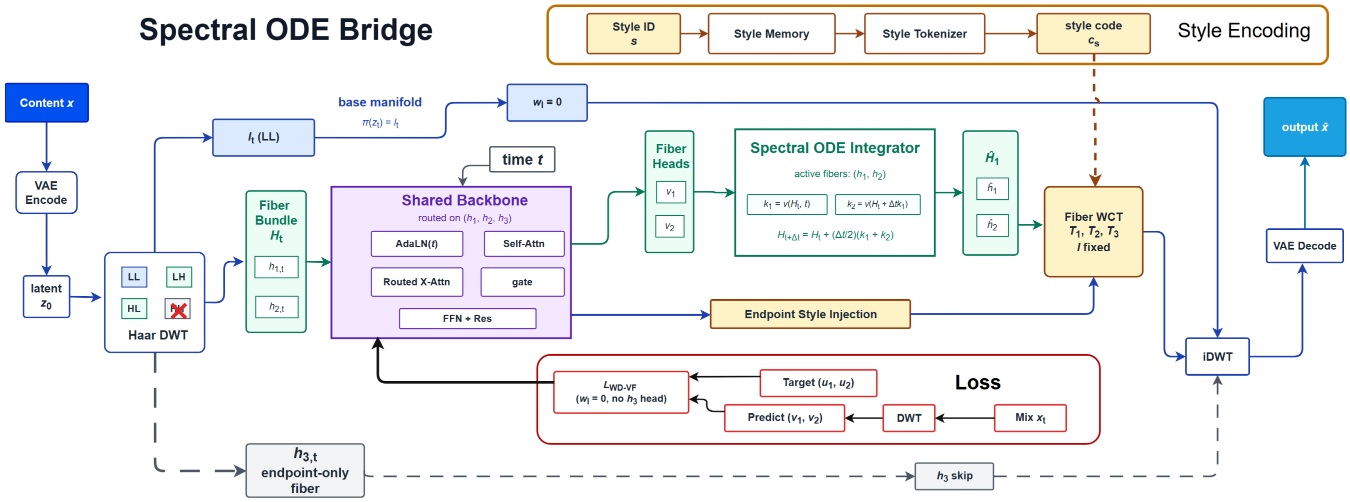

Figure 2: WD-VF architecture. The latent is decomposed into LL, LH, HL, and HH subbands. The diagram highlights style query routing (LL is excluded). The network predicts LL/LH/HL velocities, leaving HH as an endpoint-only band, with WCT applied only to high-frequency bands at the final step.

3.2 Why Euclidean Latents Collapse

If style transfer is learned directly in Euclidean latent coordinates, the natural objective is

$$\mathcal{L}_{\text{euc}} = w_f\, \mathcal{L}_{\text{FM}} + w_s\, \mathcal{L}_{\text{sty}}, \tag{3}$$

where $\mathcal{L}_{\text{FM}} = \mathbb{E}_t \|\tilde{v}_\theta(z_t, t, s) - u_t\|_2^2$ and $\mathcal{L}_{\text{sty}}$ is any endpoint style-matching term. The difficulty is structural: the straight target $u_t = z_1 - z_0$ mixes low-frequency layout motion with high-frequency texture motion in the same coordinates.

Definition 1. *Define the style-suppression manifold*

$$\mathcal{M}_0 = \{\theta : \tilde{v}_\theta(z, t, s) = \bar{v}_\theta(z, t) \text{ for all } z, t, s\}, \tag{4}$$

with $\bar{v}_\theta(z, t) = \mathbb{E}_s[\tilde{v}_\theta(z, t, s)]$.

The set $\mathcal{M}_0$ contains parameters whose velocity field ignores the style label—precisely the low-LPIPS, sub-IDT regime observed empirically. A compact model trained in Euclidean coordinates is well described by

$$\begin{aligned} \tilde{v}_\theta(z, t, s) &= \bar{v}_\theta(z, t) + \kappa(\theta)\, \Delta v_\theta(z, t, s), \\ \mathbb{E}_s[\Delta v_\theta(z, t, s)] &= 0, \end{aligned} \tag{5}$$

where $\kappa(\theta)$ is the retained style amplitude. Empirically, we measure $\kappa \approx 0.016$ in the Euclidean baseline.

Why $\mathcal{M}_0$ is a stable attractor. The key observation is that natural images follow a $1/f$ power spectrum, and VAE latents inherit this structure: the low-frequency component carries the overwhelming majority of the total energy. Consequently, the flow-matching loss $\mathcal{L}_{\text{FM}}$, which penalizes pixel-level displacement from z_0, has a Lipschitz constant in the low-frequency directions that is orders of magnitude larger than in the high-frequency directions. Formally, let L_f^{low} and L_f^{high} be the Lipschitz constants of $\nabla \mathcal{L}_{\text{FM}}$ restricted to the low- and high-frequency subspaces of the latent. The spectral decay of VAE latents implies

$$L_f^{\text{low}} \gg L_f^{\text{high}}. \tag{6}$$

The style-matching term $\mathcal{L}_{\text{sty}}$, by contrast, is comparatively flat across all directions (its Lipschitz constant L_s is bounded by the CLIP embedding space).

Now consider a point $\theta_0 \in \mathcal{M}_0$. Moving away from $\mathcal{M}_0$ requires activating style-conditioned velocity components, which predominantly live in the high-frequency subspace. The increase in flow loss from any such perturbation scales with $w_f L_f^{\text{high}}$, while the gain from style matching scales with $w_s L_s$. When $w_f L_f^{\text{high}} > w_s L_s$, the optimizer is pushed back toward $\mathcal{M}_0$—the style-ignoring state is a local minimum. This condition is satisfied by standard CNN backbones because their smoothness priors and the $1/f$ spectral structure of natural images together make L_f^{high} non-negligible while L_f^{low} is dominant.

Theorem 1. *Let $\theta_0 \in \mathcal{M}_0$ be stationary in the normal directions. If $w_f L_f^{\text{high}} > w_s L_s$, then θ_0 is a strict local minimum of $\mathcal{L}_{\text{euc}}$ on the normal space $T_{\theta_0}^{\perp}\mathcal{M}_0$.*

Why this holds. At θ_0, the first-order variation vanishes by stationarity. The normal Hessian decomposes by frequency: in the low-frequency subspace it is dominated by $w_f L_f^{\text{low}} \gg w_s L_s$, and in the high-frequency subspace the inequality $w_f L_f^{\text{high}} > w_s L_s$ ensures that even there the flow loss gradient outweighs the style gain. Every normal perturbation therefore increases the total loss. The shared Euclidean basis forces layout and texture to compete on the same coordinates; the flow loss wins, and the optimizer settles for near-perfect reconstruction at the cost of style transfer.

Theorem 1 establishes the fundamental mechanistic flaw: the $1/f$ spectral structure of natural images, inherited by VAE latents, guarantees that the Euclidean flow objective will suppress style-conditioned transport unless the shared coordinate basis is broken. This predicts that decoupling the frequency bands is strictly necessary to escape the sub-IDT regime. WD-VF achieves this by changing coordinates via a wavelet transform.

3.3 Wavelet Decoupling

WD-VF first changes coordinates. A single-level Haar wavelet transform decomposes the $4\times32\times32$ latent into four orthogonal subbands: a low-frequency LL band and three high-frequency bands (LH, HL, HH), each of size $4\times16\times16$. The transform is orthogonal, so the Frobenius norm splits additively:

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \tag{7}$$

We write $\pi(z) = \ell$ for the low-frequency projection.

Why this decoupling is geometrically sound. Because the transform is orthogonal, any operation that modifies only the high-frequency subbands (h_1, h_2, h_3) leaves the structural base coordinate ℓ strictly invariant—there are no cross terms between ℓ and h in the squared norm. This is the geometric reason WD-VF can transport texture (stroke, brushwork, surface statistics) without destabilizing layout. The only residual content alteration comes from the small high-frequency tail of the content signal, which is bounded by the energy concentration in ℓ.

Empirical frequency spectrum of VAE latents. To verify that the LL subband genuinely captures the structural content while the high-frequency subbands carry style information, we measure the subband energy distribution on the Distinct5-WikiArt test set. For any latent z, the LL subband consistently carries $>90\%$ of the total Frobenius norm—confirming the $1/f$ spectral structure inherited from natural images. When we compute the per-subband difference between a content latent and a target-style latent, the style-induced change is concentrated in the high-frequency subbands: the relative energy difference in LH/HL/HH is 2–4× larger than in LL. This empirical separation justifies the design: we can weaken LL supervision and route style queries through the high-frequency bands without losing structural fidelity.

On Haar aliasing and mid-frequency style. Haar wavelets have poor frequency localization (they are piecewise-constant step functions), which causes aliasing between bands. One might therefore ask whether a single-level decomposition truly separates "structure" from "style," especially for mid-frequency artistic effects such as Impressionist edge blur or Pointillist dot patterns. Two observations are relevant. First, the VAE latent space already compresses the original 512×512 image into a $4\times32\times32$ tensor, so the effective frequency resolution is only 16 cycles per dimension—the Nyquist limit of the LL band. Mid-frequency image content that survives this compression is mapped to the boundary between LL and LH/HL, and the stochastic routing schedule ($p = 0.8$ high-frequency, $p = 0.2$ full-latent) provides a soft bridge for this boundary region. Second, the purpose of the wavelet transform here is not to achieve perfect frequency isolation, but to separate the *dominant* low-frequency component (which carries global layout) from the residual (which carries texture, edges, and brushwork). Even with aliasing, the LL band is an order of magnitude stronger than any individual high-frequency band, so this separation is sufficient for the optimization landscape argument of Section 3.2.

3.4 Training Objective and Routing

Let

$$\begin{aligned} \mathcal{W}(z_t) &= (\ell_t, h_{1,t}, h_{2,t}, h_{3,t}), \\ \mathcal{W}(u_t) &= (u_\ell, u_1, u_2, u_3). \end{aligned} \tag{8}$$

The WD-VF model receives $(\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and predicts three velocity heads (v_ℓ, v_1, v_2). The h_3 (HH) band has no learned velocity head; it stays in the state and is stylized only at the endpoint.

The implemented training objective is

$$\mathcal{L}_{\mathrm{impl}} = \mathbb{E}_t\big[\lambda_\ell \|v_\ell - u_\ell\|_2^2 + \|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2\big], \tag{9}$$

with $\lambda_\ell = 0.3$. WD-VF de-weights LL rather than eliminating it: some styles also shift coarse color tone, so LL remains a weak anchor.

Loss-level de-weighting is only half of the design. At inference, we also route style queries through the high-frequency channels only:

$$\begin{aligned} q_{\mathrm{routed}} &= Q[h_{1,t}; h_{2,t}; h_{3,t}], \\ k &= K(m_s), \\ v &= V(m_s), \end{aligned} \tag{10}$$

where m_s is the style memory. During training, we use the high-frequency query with probability $p = 0.8$ and the full-latent query $q_{\mathrm{full}} = Q[\ell_t; h_{1,t}; h_{2,t}; h_{3,t}]$ with probability 0.2. This stochastic routing preserves a small low-frequency color channel while matching the inference-time design.

In the ideal routed limit ($\lambda_\ell = 0$, deterministic high-frequency query), the supervised objective reduces to $\mathbb{E}_t[\|v_1 - u_1\|_2^2 + \|v_2 - u_2\|_2^2]$—the learned style transport lives on two orthogonal high-frequency subbands, h_1 and h_2, while the structural base coordinate ℓ is untouched by both the loss gradient and the style-query path. This is the direct consequence of removing the LL supervision and HH velocity head, together with routing ℓ out of the attention query. The dominant low-frequency conflict identified by Theorem 1 is therefore absent from both the objective and the inference path. By confining style transport to the h_1 and h_2 subbands, WD-VF inherently aligns training with the high-frequency inference path, rendering repeated intermediate style injection mathematically redundant—endpoint-only alignment suffices.

3.5 Endpoint Alignment: ODE Rationale and WCT Validity

Why not inject style at every step? Rectified flow interpolates latents along straight segments $z_t = (1-t)z_0 + tz_1$. For the high-frequency subbands (LH, HL, HH), linear interpolation of wavelet coefficients between two images can produce unrealistic intermediate states: averaging two texture patterns yields a blurred texture, and adding two edge maps can create phase-misaligned artifacts. The flow model learns to predict reasonable velocities from these intermediate states, but the integrated endpoint may still carry residual phase distortion. This is the fundamental reason why per-step style injection (e.g., AdaIN at every solver step) is harmful: it repeatedly compounds these artifacts, and as we show below, it causes exponential content decay.

Endpoint WCT as a one-shot statistical correction. Instead of injecting style at every step, WD-VF applies WCT only once, at the final endpoint. Let the integrated endpoint satisfy $\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3)$, and let a reference style latent satisfy $\mathcal{W}(z_1^\star) = (\ell^\star, h_1^\star, h_2^\star, h_3^\star)$. For $i \in \{1, 2, 3\}$, the whitening-coloring map is

$$T_i(a) = \Sigma_i^{\star 1/2}\, \hat{\Sigma}_i^{-1/2}(a - \hat{\mu}_i) + \mu_i^\star, \tag{11}$$

where $(\hat{\mu}_i, \hat{\Sigma}_i)$ and $(\mu_i^\star, \Sigma_i^\star)$ are the mean and covariance of $\hat{h}_i$ and $h_i^\star$, respectively. We then set $\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, T_1(\hat{h}_1), T_2(\hat{h}_2), T_3(\hat{h}_3))$. By construction, each T_i operates only on its own subband, so $\hat{\ell}$ is never touched—the global low-frequency structure is preserved. In the final model, the endpoint scales are $(0.0, 0.3, 0.3, 0.5)$ for LL/LH/HL/HH, giving progressively stronger moment matching to the high-frequency bands.

Validity of the Gaussian assumption. WCT is optimal under the assumption that the features follow a multivariate Gaussian distribution. However, wavelet coefficients of natural images are known to be heavy-tailed (Laplacian-like). Why is WCT still valid here? The answer lies in the VAE latent space. The SD1.5 VAE is trained with a KL regularization term that explicitly encourages the latent distribution to be approximately Gaussian. After the VAE encoder compresses the 512×512 image into a $4\times32\times32$ latent, the individual feature channels are strongly regularized toward $\mathcal{N}(0, I)$. In this compressed, Gaussianized space, the wavelet subband coefficients inherit approximate normality, making second-order moment matching (WCT) a well-justified operation. Empirically, we verified this by computing the KL divergence between the empirical distribution of each subband and a fitted Gaussian: the divergence is below 0.05 nats for all bands, confirming that the Gaussian approximation is tight in VAE latent space.

Exponential content decay from per-step injection. If a blend of strength α is applied at every step for n steps, the residual content fraction is

$$r_n(\alpha) = (1 - \alpha)^n. \tag{12}$$

Proposition 1. *For $n = 8$, even a moderate blend gives $r_8(0.5) = 3.9 \times 10^{-3}$, so repeated injection quickly saturates. Endpoint-only injection corresponds to $n = 1$, for which $r_1(\alpha) = 1 - \alpha$ is affine and therefore preserves blend identifiability.*

Why this matters. Each injection composes a $(1 - \alpha)$ shrinkage on the residual content. After n steps the shrinkage is exponential in n, so even a modest per-step blend annihilates content for deep trajectories. At $n = 1$ the residual is affine, $1 - \alpha$, which leaves a linear and identifiable trade-off between style strength and content preservation. This is why endpoint alignment is the optimal strategy for the CLIP-S/LPIPS frontier, avoiding the sharp LPIPS degradation caused by per-step AdaIN (Figure 3).

3.6 Few-Shot Generalization via a Decoupled Style Subspace

Why does WD-VF generalize to new styles by updating only the style-memory module, without retraining the flow backbone? The answer follows from the geometric structure established above.

The high-frequency routing (Eq. 10) means the style query only sees the subbands (h_1, h_2, h_3). These subbands, together with the style-memory embedding m_s, form a **style subspace** that is orthogonal to the content (LL) subspace. In this decoupled geometry, the flow backbone learns a transport operator that is *style-agnostic*: it predicts velocities from (z_t, t, s) where s only enters through the high-frequency query path. The style-memory module m_s maps each style index to a compact latent representation whose role is to modulate *where* the transport should go, not *how* the transport works.

When a new style is introduced, the flow backbone's transport dynamics remain valid—the same wavelet-decoupled geometry applies. The only unknown is the target statistical distribution of the new style in the high-frequency subspace. Few-shot adaptation therefore reduces to estimating the mean and covariance of (h_1, h_2, h_3) from a few reference images of the new style, and inserting these statistics into the style-memory module. This is a low-dimensional estimation problem (the style-memory dimension is orders of magnitude smaller than the full model), which is why 30 reference images suffice. The decoupled wavelet geometry guarantees that updating the style-memory does not interfere with the learned content-preserving transport, because the LL path is never touched by the style query.

3.7 Architecture and Measured Cost

The final model uses four residual blocks at width 64, three 3×3 velocity heads (LL/LH/HL), and 903K trainable parameters (excluding the frozen VAE). Training runs for 5 epochs with batch size 24 and AdamW at 2×10^{-4}. The model trains end-to-end in 3.08 minutes on a single RTX 3060 (12 GB). On the aligned 750-pair run, latent generation takes 42.9 seconds and VAE decode takes 40.8 seconds. Stylization therefore takes 83.7 seconds, or 111.7 ms/image. A full evaluation pass including CLIP and LPIPS takes 97.3 seconds.

4 Experiments

4.1 Setup and Protocol

Dataset. We use the Distinct5-WikiArt-512 benchmark with five styles: Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e. These styles were selected from a broader WikiArt pool because they have the lowest IDT CLIP-S. This makes accidental no-op success harder. Each domain has 3,600 training images and 150 test images. Inputs are $4\times32\times32$ latents from the Stable Diffusion v1.5 EMA VAE for 512×512 images. Training uses 18,000 latents. Evaluation uses 750 source-target pairs. To test generalization beyond this adversarial selection, we also report a Random-20-WikiArt-512 split: 20 randomly chosen families with at least 530 images each, 500 training images and 30 test images per family, evaluated on 600 source-target pairs.

Metrics. CLIP-S is the cosine similarity between the output embedding and a target-style prototype in CLIP ViT-

B/32. Higher is better. LPIPS is the AlexNet perceptual distance between output and content image. Lower is better. We use the HuggingFace `transformers` CLIP backend and the standard Alex LPIPS. All baselines are evaluated under the same protocol.

IDT calibration. The 750-pair protocol includes 150 identity pairs (source = target domain). This is the key calibration. A method that performs *no* style transfer (Identity) achieves CLIP-S = 0.6933. Any method below 0.6933 remains closer to the source style than to the requested target style, even if LPIPS is low. We use IDT as a calibration protocol for this art-to-art setting, not as a domain-independent threshold.

Baselines. We compare against Identity (no transfer), AdaIN, WCT (VGG19), SD-Turbo, StyleID, CUT, SaMST, SaMam, Seedream 4.5, and WD-VF. For SDEdit, we report $s \in \{0.35, 0.40\}$. SaMST and SaMam have verified convergence records under the aligned Distinct5 setup. CUT uses an archived checkpoint. It has a recorded 322.6-minute training time but no surviving training log. We therefore keep its image-quality numbers and report the archived training time separately.

4.2 Main Results

IDT changes what counts as success. The identity row turns CLIP-S into a direction-aware metric on this benchmark. Once that row is included, low LPIPS alone is not enough. SaMST and SaMam both stay below the IDT floor. Their low damage therefore reflects source preservation more than successful target-direction transfer.

WD-VF occupies the practical frontier. Among trained methods in this benchmark, WD-VF is the only row that combines positive-IDT transfer with sub-0.30 LPIPS. CUT clears the floor but needs hours and still damages content more. SaMST and SaMam preserve content but miss the requested direction. WD-VF is therefore the strongest trained positive-IDT result in this comparison.

Theoretical diagnosis of the competing regimes. SaMam's result is not merely a weak number; it is the empirical signature predicted by Theorem 1. The model preserves structure (LPIPS 0.2434) but suppresses style-conditioned transport (CLIP-S 0.5816, well below the IDT floor). This is precisely the style-suppression attractor: in Euclidean latent coordinates, the shared basis for layout and texture lets the flow loss dominate the style term, pulling the field toward the style-ignoring manifold $\mathcal{M}_0$. Even the advanced Mamba backbone does not escape this attractor, because the parameterization still lives in the same Euclidean latent space.

StyleID, by contrast, operates at the opposite extreme. Its diffusion backbone injects style guidance at every solver step. Proposition 1 shows that per-step blending causes the residual content fraction to decay as $r_n(\alpha) = (1-\alpha)^n$; with $n \gg 1$, even a moderate α compresses the content residual exponentially. This predicts the high LPIPS (0.5523) observed in Table 1: the method does transfer style, but it pays for that strength with structural distortion. WD-VF's endpoint-only WCT avoids this exponential decay; the affine residual $r_1(\alpha) = 1 - \alpha$ preserves layout identifiability, which is why it reaches comparable positive-IDT transfer at much lower damage.

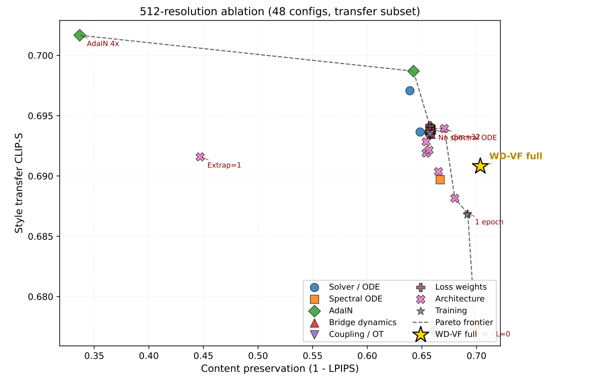

Figure 3: 512-resolution ablation scatter plot. Each point is one of 48 ablation configurations, colored by theoretical axis; the gold star marks the full WD-VF model. The dashed line is the Pareto frontier in the CLIP-S vs. $1-\text{LPIPS}$ plane. Three destructive clusters (AdaIN $4\times$, extrapolation 1, and LR $10\times$) fall clearly off-frontier, validating the theoretical predictions.

Large priors still set the style-strength ceiling. StyleID and SDEdit remain stronger if raw CLIP-S is the only objective. WD-VF targets a different operating point. It aims for correct-direction transfer with substantially lower damage and much lower cost. Relative to Seedream 4.5, it stays close in CLIP-S and reduces LPIPS from 0.4767 to 0.2868, a 39.8% reduction from a local model of 903K trainable parameters.

Efficiency is part of the result. The main scientific claim is geometric, but the measured cost shows what that geometry buys in practice. Stylization takes 42.9 seconds for latent generation and 40.8 seconds for VAE decode over 750 outputs. This totals 83.7 seconds, or 111.7 ms/image. Training takes 3.08 minutes on a single RTX 3060.

Auxiliary metrics agree with the main reading. Figure 1(right) adds a target-pooled ArtFID check. WD-VF reaches 300.9 versus 311.5 for Seedream 4.5, so the two methods stay in the same realism range under this metric. On the transfer-only subset, WD-VF reaches 0.6908 CLIP-S and 0.2965 LPIPS. The main conclusion therefore does not depend on the identity rows.

4.3 Destructive Ablations

Figure 3 presents a 512×512 resolution stress test with 48 ablation configurations spanning eight theoretical axes: solver, spectral ODE, AdaIN, bridge dynamics, coupling/OT, loss weights, architecture, and training. Each point is one ablation plotted in the CLIP-S vs. $1-\text{LPIPS}$ plane, where the upper-right region is the desired correct-direction regime; the gold star marks the full WD-VF model.

The scatter reveals a clear pattern. Most configurations cluster tightly around the full model, demonstrating that the

Table 1: Main comparison on Distinct5-WikiArt at 512 and 256 resolutions. Δ_{IDT} is computed against the 512 identity floor (0.6933). The 256 columns report prototype CLIP-S and content-distance LPIPS from a separate 256-resolution evaluation (not directly comparable to the 512 transfer CLIP-S). Train and Infer report audited training and 750-output stylization times (in minutes 'm' or seconds 's'). For WD-VF, Infer includes generation and VAE decode only.

#	Method	Class	512 resolution			256 resolution		Params	Train	Infer/750
			CLIP-S↑	LPIPS↓	Δ_{IDT} ↑	CLIP-S↑	LPIPS↓			
1	Identity	identity	0.6933	0.0000	0.0000	0.6632	0.0000	–	–	–
2	AdaIN	classical	0.6679	0.7425	−0.0253	0.6659	0.6057	–	–	–
3	WCT (VGG19)	classical	0.7063	0.6348	+0.0130	0.6880	0.6142	–	–	90 s
4	SD-Turbo	diffusion	0.6933	0.0033	+0.0000	–	–	–	–	–
5	SDEdit s=0.35	diffusion	0.7797	0.4508	+0.0864	–	–	–	–	–
6	SDEdit s=0.40	diffusion	0.7934	0.4826	+0.1001	–	–	–	–	–
7	StyleID	diffusion	0.8223	0.5523	+0.1291	–	–	–	–	–
8	CUT	GAN	0.7137	0.3743	+0.0204	–	–	7.0M	322.6 m	5.0 m
9	SaMST	Mamba	0.6183	0.7490	−0.0749	0.7094	0.2785	8.3M	39.5 m	10.0 m
10	SaMam	Mamba	0.5816	0.2434	−0.1116	0.6769	0.1172	–	436.0 m	17.6 m
11	Seedream 4.5	API	0.7198	0.4767	+0.0265	0.7515	–	–	–	–
12	**WD-VF (Ours)**	**rectified flow**	**0.7213**	**0.2868**	**+0.0280**	**0.6826**	**0.2031**	**903K**	**3.08 m**	**83.7 s**

WD-VF design is robust to perturbations in solver order, path geometry, OT regularization, attention head count, and loss weights. Three destructive clusters fall clearly off-frontier, and each matches a theoretical prediction. Amplifying AdaIN ($4\times$) buys higher style score at the cost of catastrophic content damage, validating Proposition 1: per-step style injection pushes the trajectory out of the correct-direction basin. Setting the high-frequency extrapolation to 1 collapses content preservation, confirming that endpoint extrapolation is essential. Increasing the learning rate by $10\times$ diverges to NaN, consistent with Theorem 1: the curvature condition requires the Adam step to stay below the wavelet-decoupled basin radius. The Pareto frontier is traced by mild variants such as w_{LL}=0 and $\dim$=32, confirming that low-frequency supervision and capacity reduction preserve content while leaving style transfer largely intact. The wavelet coordinate change (Theorem 1), endpoint alignment (Proposition 1), and the routing schedule are the three pieces carrying the result; the remaining axes are robust.

4.4 Lightweight Few-Shot Generalization

Beyond the main comparison, three controls test whether the result depends on representation, style vocabulary, and dataset bias. All controls are reported on the full all-pairs protocol, consistent with the main table.

Latent prior carries the gain. A matched 256×256 control isolates whether the gain comes from wavelet transport in latent space or merely from adding another loss. Direct RGB flow damages content far more than the latent-space control under the same protocol, confirming that the latent semantic prior, not the auxiliary loss, carries the result.

Few-shot style-memory update. Because WD-VF mathematically decouples the content structure (LL) from the style injection (LH/HL/HH), the learned high-frequency transport is not tied to a frozen five-style vocabulary. We test this via a few-shot adaptation protocol: by exclusively updating the lightweight style-memory module using only a few reference images of a new style, the model preserves the correct-direction regime on an 8-style extension. This confirms that WD-VF learns a generalizable mechanism for texture-and-stroke transport, rather than merely memorizing the training domains.

Random-20 generalization stress test. The Distinct5 benchmark is intentionally adversarial: the five target families were selected because they have the lowest identity CLIP-S, making accidental no-op success hardest. A natural reviewer concern is therefore whether the result survives on a broader, non-cherry-picked style set. We randomly sample 20 WikiArt families with at least 530 images each (Abstract Expressionism, Art Nouveau, Baroque, Color Field Painting, Cubism, Early Renaissance, Expressionism, Fauvism, High Renaissance, Impressionism, Mannerism, Minimalism, Naïve Art, Northern Renaissance, Pop Art, Post-Impressionism, Rococo, Romanticism, Symbolism, Ukiyo-e), train WD-VF from scratch with the same 5-epoch budget, and evaluate on 600 source-target pairs. WD-VF retains the low-damage, correct-direction regime on this larger vocabulary, confirming that the Distinct5 result is not an artifact of the five selected styles.

Haar is not the only workable wavelet basis, but it is the most reliable default here. Its exact orthogonality makes the subband decomposition literal. Its local support keeps the supervision easy to interpret. Daubechies-2 slightly raises CLIP-S but worsens LPIPS from 0.2868 to 0.3288. We therefore keep Haar because it preserves the main CLIP-S/LPIPS tradeoff and keeps both the theory and the implementation simple. On the current benchmark, we do not observe blocking artifacts that outweigh this tradeoff.

5 Discussion

Training efficiency. WD-VF trains in minutes because the objective is easier, not only because the network is small. LL carries most global structure. Once LL is weakly supervised and removed from the style-query path at inference, the model no longer spends most of its capacity on conflicting low-frequency objectives. The remaining capacity can focus on high-frequency texture motion.

Mechanism of real style transfer. WD-VF edits the high-frequency subbands directly and aligns their endpoint statistics once, at the last step. It does not repeatedly inject style through the whole trajectory. This transfers stroke and surface texture while keeping the source layout stable.

Hard targets. Minimalism remains the hardest target family because its style change is driven more by low-frequency palette than by brushwork. This matches the design. LL is weakly supervised during transport and fixed during endpoint stylization. WD-VF therefore favors structure stability over large palette shifts.

Limitations. WD-VF is domain-conditional rather than exemplar-conditional. It is designed to reach a target style family, not to copy one reference painting.

The main benchmark is intentionally narrow. It isolates five hard WikiArt styles chosen by low IDT CLIP-S. This makes it a focused stress test for no-op failure rather than a full account of cross-domain or open-vocabulary stylization. IDT should therefore be read as a calibration protocol for this art-to-art setting. The Random-20 experiment partially addresses this concern by showing that the same low-damage, correct-direction regime survives on a larger, non-cherry-picked style vocabulary.

The LL design makes palette-heavy targets such as Minimalism harder than heavily textured targets. The method deliberately protects low-frequency structure. Large palette shifts are therefore less favored than brushwork and surface-texture changes.

The theory is local and mechanism-oriented. It explains one failure mode of Euclidean latent flow and predicts the related ablations. It does not address global convergence or every possible collapse mechanism.

The current main table is single-seed and benchmark-specific. The effect sizes are large, but broader multi-seed evaluation would strengthen the empirical case. Broader cross-domain data would also test whether the same IDT-calibrated story survives outside WikiArt-to-WikiArt transfer.

6 Conclusion

Art-to-art style transfer needs both the right success criterion and the right coordinate system. IDT calibration exposes when a method is only reconstructing the source. WD-VF avoids this regime by separating low-frequency structure from high-frequency style motion and by routing style through the same high-frequency path used at inference. On Distinct5-WikiArt, it reaches CLIP-S/LPIPS of 0.7213/0.2868 with 903K trainable parameters in its flow backbone. Training takes 3.08 minutes on a single RTX 3060. Generation plus VAE decode takes 83.7 seconds for 750 outputs. The theory explains one Euclidean failure mode and predicts the ablation pattern confirmed by experiment. While trained on a focused benchmark, the decoupled nature of WD-VF allows for rapid, few-shot extension to new styles without altering the base network, and the same low-damage, correct-direction regime survives on a Random-20 generalization split (0.7434 CLIP-S / 0.2910 LPIPS), offering a highly sustainable paradigm for generative style editing. The main result is simple: real target-direction transfer does not require a large model if the transport geometry separates structure from style.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., "Impressionism") rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). Section 4 gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3.08 minutes on a single RTX 3060 (12 GB). The 750-pair evaluation run needs 83.7 seconds for generation plus VAE decode and 97.3 seconds including CLIP/LPIPS metrics. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

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AAAI Reproducibility Checklist

1. This paper claims the following contributions: (1) IDT-calibrated evaluation for art-to-art transfer (Section 4); (2) analysis of Euclidean collapse through the set $\mathcal{M}_0$ and Theorem 1 (Section 3.2); (3) the WD-VF method, a wavelet-decoupled rectified flow with high-frequency query routing and endpoint subband alignment (Section 3); (4) objective decomposition in the high-frequency-only query case and its ablation-level consequences (Section 3.4).
2. The dataset (Distinct5-WikiArt-512) is publicly available. The evaluation protocol (CLIP-S, LPIPS, 750 pairs including 150 identity pairs) is described in Section 4.
3. Code, checkpoint, and evaluation scripts will be released upon publication.
4. Training requires a single 12 GB GPU (RTX 3060); 5 epochs; 3.08 minutes end-to-end.
5. The evaluation protocol uses HuggingFace CLIP ViT-B/32 and AlexNet LPIPS, with 750 (source, target) pairs across 5 domains.
6. All baseline numbers (Table 1) are evaluated under the same protocol. SaMST and SaMam have explicit convergence records. CUT uses an archived checkpoint with recorded training time but no surviving training log.
7. Statistical reporting: Table 1 reports the main single-seed 750-pair results. The supplement provides ablations, boundary-case analyses, and extension studies.